Compressed Saturated Interval Bundles and Exact Full-Row Cancellation

A source-certified advance toward efficient unknot recognition

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Abstract

We extend the source-certified normal-cut handlebody consumer to a class of fully attached bordered interval bundles. A saturated base contributes one residual two-handle. We prove that its incidences with the original core cocores are the unsigned sums of the authenticated attaching-annulus incidences; its new private cocores are each met twice. The private columns can never become unit pivots, so an arbitrarily large private handle count can remain a binary integer. A connector-forest correction gives the exact genus when several original cores are joined.

For the resulting rectangular nonnegative matrix, full positive cancellation is equivalent to immutable unit-column peeling, a permanent-one full-row minor, and absence of a nonempty stopping subset. The matching criterion is a consequence of existing ProveIt report 56; the new implementation gives linear incidence processing and an independent timestamp verifier. We also prove that the maintained minimum-fill planner already decides full-row completion correctly, correcting its stated scope. An optional early core filter follows from row-subset monotonicity.

On the pinned 106-case native corpus, successful handlebody censuses increase from 78 to 85, preserving all old successes. All 96 compiled conclusions independently replay; 78 positive results match literal Regina cuts, while seven binary-scale inputs exceed the explicit expansion cap. The remaining 21 cases stay inconclusive. A separate ambient-surface repair theorem gives an additive potential for a precisely specified class of seam-free backward repairs. We provide proofs, code, adversarial tests, preserved measurements, and a research program. These are local advances; the missing bounds on surface discovery, geometric transport, and global hierarchy search are identified explicitly.

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1 Results, provenance, and the exact target

The concrete advance in this continuation is a geometric compiler. It turns an authenticated normal cut with saturated bordered interval bundles into a small incidence matrix and a binary count of private one-handles. Complete cancellation of that matrix proves an exact handlebody census of the entire supplied cut. This resolves a specific limitation of the maintained consumer: a fully attached Möbius, annulus, or higher-genus bundle need no longer be rejected merely because its boundary is saturated. Some such inputs still fail the geometric cancellation criterion, as the examples and native audit demonstrate.

The companion algebraic improvement removes repeated mutable-matrix scans. It also replaces a stalled search report by a short, canonical certificate of impossibility *within the fixed positive-incidence model*. Both the gain and the restriction matter. Eight native raw obstructions in our audit are cuts that Regina identifies as unions of handlebodies. Their failure points to missing geometric simplifications, rather than an error in the negative matrix certificate.

1.1 Repository position and contribution boundaries

We inspected the repository and incoming directory at commit `704600626606` (full identifier on the title page), dated 12 October 2026[1]. The executable comparison uses that exact maintained `fastunknot` source. The inherited corpus was originally recorded at revision `d2c5e1707fe7`; we reran its 106 inputs against the inspected implementation. The article does not equate an archive's original baseline with the current baseline.

Existing source	Relevant result already available	Advance made here
Report 56[2]	Raw positive elimination, permanent-one matching, and source closure	Rectangular core consumer; column peeling; canonical obstruction; independent sparse replay
Report 85 and synthesis[3]	Actual cut geometry, ordered source ports, disk-bundle absorption, geometric cancellation	A residual incidence row for every saturated bordered base
Incoming hierarchy kernels[4]	Relative band counts; unsaturated bundles; one residual two-handle for a saturated base	Exact source-core incidence of that residual; private-column omission; a checked whole-cut census
Incoming disconnected tiles, sparse cut grammars, and moment-coded first returns[5, 6, 7]	Compressed local kernels and small-query boundary tools	A geometric consumer for a different outstanding interface; their abstract kernel results are not claimed again

Table 1: Relation to the material already present in the two requested repository directories. The novelty claim is local to the stated source interface and implementation, not a claim of priority for classical handle theory or unique-matching arguments.

The relative handle count itself is therefore recalled as a building block. The new mathematical step is to determine which old cocores the saturated residual curve meets, while proving that all its newly introduced columns are inert. This requires actual product charts. Euler characteristics and abstract relator words do not imply the needed statement. The source-chart condition, and the collar isotopy fixed on the actual gluing boundary, are made explicit in Section 3.

The additive repair theorem later in the article addresses a second, global-direction question:

when can a backward correction preserve other surfaces instead of discarding them? It applies to a fixed, authenticated ambient block, and establishes neither an implementation nor a universal replacement for the hierarchy reset rule.

1.2 Which quasi-polynomial announcement?

The two author-hosted 2021 slide versions give different exponents. Their distinction should be retained in the project rather than silently resolved by choosing one file.

Primary source	Announced bound for an n-crossing diagram
February 2021 UC Davis slides, PDF pp. 13–14[8]	$n^{O(\log n)} = 2^{O((\log n)^2)}$
March 2021 Oxford slides, PDF pp. 13–14[9]	$2^{O((\log n)^3)} = n^{O((\log n)^2)}$

Table 2: Both bounds are quasi-polynomial. The February target in the repository has a genuine primary-source antecedent.

Lackenby's July 2026 paper[10] proves the hierarchy algorithm and substantial certification results. Section 9 leaves some operation costs unspecified; Proposition 9.1 bounds step iterations by $L(g+1)^L$. Proposition 10.2 bounds the refined hierarchy length by $4c_q(\mathcal{H})$. These statements do not, by themselves, prove either slide running time. Our composition analysis in Section 4 explains exactly which additional size, discovery, and depth bounds would suffice.

1.3 Four distinct mathematical outputs

We keep the following conclusions distinct throughout.

1. **A valid source decomposition.** The supplied triangulation and normal coordinates determine actual complementary core handles and embedded interval bundles, with authenticated attachment order.
2. **A compiled relative presentation.** The old core rows, the saturated residual rows, and the private handle counts describe a genuine handle structure under the source-chart contract.
3. **A complete cancellation or a raw obstruction.** A complete positive cancellation proves that the associated cut components are handlebodies. A stopping subset proves that the particular cancellation model cannot delete every row.
4. **A knot verdict.** This needs the separate, established topological reduction from the knot exterior and its selected surfaces. The new whole-cut operator returns no standalone unknot or knottedness declaration.

This separation is mathematical. For example, a cut of a trefoil exterior can itself be a handlebody; discovering that fact alone cannot establish that the original knot is an unknot. Likewise, a component with Euler characteristic one and spherical boundary need not be a ball. The certificate must account for its actual handles.

2 The exact rectangular cancellation criterion

The maintained core solver first peels rows and, for rectangular matrices, falls back to a saturated positive-pivot heuristic. There is a dual source criterion that decides full row completion directly. This is a consequence of report 56's raw-epoch theorem[2], applied to the transposed incidence matrix with an empty survivor alphabet. We give a direct proof because it reveals a short

obstruction certificate and avoids searching over surviving columns. No priority claim is made for the underlying matching or closure criterion.

Let $A = (a_{ic})$ be a nonnegative integral $m \times n$ matrix. A *positive unit pivot* at (r, c) is allowed when $a_{rc} = 1$; it deletes row r and column c and changes every surviving entry to

$$a'_{ih} = a_{ih} + a_{ic}a_{rh}.$$

The question is whether some sequence deletes every row. All occurrences are unsigned geometric incidences; signed exponent sums and freely reduced words define a different problem.

A *column peel* deletes row r using a column c whose total over the live rows is exactly one. Then $a_{rc} = 1$ and $a_{ic} = 0$ for $i \neq r$, so the positive update has no fill. A set U of source rows is a *stopping set* when

$$\sum_{i \in U} a_{ic} \neq 1 \qquad \text{for every column } c.$$

The empty set is allowed, and a nonempty zero row is itself an obstruction.

Lemma 2.1 (Reverse unit-pivot lemma, recalled with proof)**.** *Let A be square and let B result from a positive unit pivot. If* $\operatorname{per}(B) = 1$*, then* $\operatorname{per}(A) = 1$*.*

Proof. Regard each entry as the number of separately labelled edges between its row and column. An edge counted by b_{ih} has a tag indicating either an original edge counted by a_{ih}, or an ordered pair consisting of an a_{ic} edge and an a_{rh} edge. Call the latter a composite edge.

The unique tagged perfect matching of B has at most one composite edge. Otherwise, take two such edges (i, h) and (j, k). Exchanging their second factors gives valid composite edges (i, k) and (j, h), and hence a distinct perfect matching. If there is no composite edge, append the unique pivot edge (r, c) to obtain a matching of A. If there is one, replace it by its two labelled source edges. This again produces a matching of A.

There is also an injection from perfect matchings of A to tagged perfect matchings of B. A matching using (r, c) maps to its restriction, with all edges tagged as original. A matching not using (r, c) has unique edges (i, c) and (r, h); replace them by their composite edge (i, h) and tag the remaining edges as original. The image, including its tags, recovers the source matching in either case. Therefore A has at most one matching, and the preceding construction shows that it has at least one. □

Theorem 2.2 (Full-row positive cancellation)**.** *The following statements are equivalent.*

1. *Some positive unit-pivot sequence deletes every row of A.*
2. *A has an $m \times m$ submatrix, using all rows, with permanent one.*
3. *Column peeling deletes every row, for every choice of eligible column.*
4. *The source has no nonempty stopping set.*

Consequently no enumeration of the $n - m$ surviving columns is required.

Proof. For $m = 0$ all statements hold with the empty minor having permanent one. If a positive-pivot sequence succeeds, record its m chosen source columns. Restrict to these columns throughout the sequence. The restricted matrices obey exactly the same positive-pivot rule and end at the empty matrix. Lemma 2.1, proved in report 56 and recalled above, implies that permanent one after a positive pivot gives permanent one before it. Reversing the complete sequence proves the second statement. In particular $m \leq n$ is necessary.

Suppose a chosen square minor has permanent one. Its unique edge-labelled perfect matching uses unit entries. For any nonempty source-row set U, restrict that matching to U and to its

matched columns. This smaller square matrix also has permanent one: another matching would extend, using the unchanged matching outside U, to another matching of the original minor. Relabel its matched entries onto the diagonal. If every column had another positive entry, the directed graph of off-diagonal positive entries would have positive indegree at every vertex and hence a directed cycle. Exchanging the matched columns around that cycle would produce a second perfect matching. Some column therefore contains just one unit entry in the rows U. Thus U is not a stopping set, proving the fourth statement.

If peeling stops on a nonempty set U, every column has total zero or at least two on U, so U is a stopping set. Under the fourth statement this cannot happen, whatever eligible columns were selected. This proves the third statement. Finally every column peel is a legal positive pivot with zero fill, proving the first statement from the third. □

Theorem 2.3 (Canonical obstruction core)**.** *Column peeling leaves exactly the union K of all source stopping sets. Thus its residual row set is independent of pivot order. A nonempty subset U satisfying the stopping inequalities is already a complete certificate that full-row positive cancellation is impossible; maximality is unnecessary for that conclusion.*

Proof. The union of stopping sets is a stopping set. Indeed, if a column has a positive entry somewhere in the union, that row lies in one constituent stopping set, where the column total is already at least two. Its total in the union is consequently at least two.

No peeling sequence can delete a row of a fixed stopping set U: at the first alleged deletion, every row of U is still present, so a column meeting the selected row has total at least two. Therefore the union K survives every peel. The final residual R is itself a stopping set, hence $R \subseteq K$. The two inclusions yield $R = K$. □

The scope of a negative answer. The all-one 2×2 matrix is a stopping core, but it has four allowed unit pivots. Any such pivot leaves the one-entry matrix (2). Column peeling therefore computes neither a maximum partial cancellation nor a minimum topological handle decomposition. An obstructed authenticated core can still be a handlebody after geometric moves outside this raw model.

Corollary 2.4 (Row-subset monotonicity and an early core filter)**.** *If a matrix permits full-row positive cancellation, so does every matrix obtained by restricting it to a subset of rows. Equivalently, adding rows cannot rescue an obstructed source matrix. This remains true if one also adds columns whose entries in all original rows are zero.*

For the saturated-bundle construction, an original core stopping subset therefore remains an obstruction in the joint presentation. With the same original-core forest fixed, the original core can be checked before constructing the full ordered bundle boundary interface. An obstruction suffices to return an inconclusive result from this particular handlebody certificate method.

Proof. Take a complete column-peeling schedule and retain only its moves on the chosen row subset. At each retained move, the chosen column has the same single unit occurrence as before: the omitted rows can only reduce its total, and the selected row itself remains. Thus the restricted schedule also completes.

Alternatively, a stopping subset of original rows remains a stopping subset after adding any other rows. Its column totals are computed only over the chosen subset, and any newly added column contributes zero there. The saturated compiler leaves original-core rows unchanged in active columns and gives them zero in private bundle columns. The same obstruction therefore applies to its joint matrix. □

This is a statement about the fixed raw geometric presentation. It is not the topological assertion that attaching another piece can never turn a non-handlebody into a handlebody. A

different original-core forest, geometric normalization, new disk system, or an allowed two/three-handle cancellation may change the raw problem and lies outside the early filter's conclusion.

2.1 Sparse discovery and independent verification

Lemma 2.5 (Columns with no unit entries are inert)**.** *Suppose every initial entry of a column is zero or at least two. That column can never contain a unit entry under any positive-pivot sequence. Removing any family of such columns leaves the answer to full-row completion unchanged. The omitted columns must still be counted when computing surviving genus.*

Proof. For an omitted column h, a surviving entry changes from a_{ih} to $a_{ih} + a_{ic}a_{rh}$. If both a_{ih} and a_{rh} are zero or at least two, their displayed sum is again zero or at least two. Induction shows that h can never be a pivot column. Updates in any other column do not depend on h, so a sequence on the full matrix projects to the same sequence after omission, and any sequence on the smaller matrix lifts with identical pivot entries. Omission is an algebraic search reduction, not deletion of the corresponding one-handles from the topological object. □

This applies in particular to private columns whose only initial entry is two, as in the saturated-bundle compiler. The invariant requires only the absence of unit entries; divisibility by two is stronger than necessary.

Replace every positive source multiplicity by its cap $\min(2, a_{ic})$. This leaves all stopping inequalities and column-singleton predicates unchanged. For each column maintain the *uncapped sum of these capped entries* over live rows, together with the xor of their distinct dense row indices. A column of total one has exactly one unit occurrence, whose row is the xor. On deleting that row, visit its immutable source incidences once, subtract their capped entries from the column totals, and xor out the row index. Enqueue a column when its total first becomes one.

Column totals decrease strictly when visited. Each column enters the queue at most once and each nonzero source incidence is removed at most once. If I is the number of nonzero source incidences, discovery uses $O(m + n + I)$ expected dictionary operations and working entries, apart from sorting the reported residual labels. That final normalization costs $O(m \log(m + 1))$ comparisons. Dense indices and capped-entry column totals have $O(\log(m + n + 1))$ bits. Arbitrary source-label bit lengths and the time to read uncapped input integers are charged separately.

The independent verifier does not simulate the queue. It assigns a deletion time to every row in the certificate and records the proposed pivot column. For pivot (r, c) at time t it checks from the immutable source that $a_{rc} = 1$ and that every other row incident to c has a deletion time less than t. A single scan of source incidences verifies these inequalities for all moves. The same scan checks that every residual column has total zero or at least two. Thus the certificate proves both a legal deletion sequence and a fixed point; Theorem 2.3 identifies the latter as the canonical obstruction. On a complete result the residual is empty.

2.2 A correction to the old producer's status description

The pinned native saturated producer minimizes

$$\Big(\sum_h a_{rh} - 1\Big)\Big(\sum_i a_{ic} - 1\Big)$$

over unit pivots, with the entries capped at two. A zero score means a unit row leaf or a unit column leaf. In fact this producer is already complete for the *full-row* decision, although its docstring describes it as incomplete.

Proposition 2.6 (Completeness of minimum-fill greedy for full completion)**.** *Any positive-pivot producer that always prefers a zero-fill pivot whenever one exists decides full-row completion correctly, provided it runs until all rows disappear or no unit pivot remains.*

Proof. Assume full completion exists. Theorem 2.2 supplies a unit column leaf, so a zero-fill pivot is available. It remains to show that either type of zero-fill pivot preserves existence of a permanent-one minor.

For a row leaf, its sole unit column belongs to every matching covering all rows. Remove that forced matched pair from a permanent-one minor. For a column leaf at row r, if the minor already includes that column the same argument applies. Otherwise replace the column matched to r by the new leaf column. Any matching of the resulting square minor must pair this column with r; another matching on the remaining rows would extend to a second matching of the old minor. Thus the replacement minor still has permanent one. Removing the leaf pair leaves a permanent-one minor for the residual matrix. Since zero-fill pivots do not alter surviving entries, induction proves completion.

Conversely a completed output is itself a positive-pivot sequence. If the source is obstructed, no choices made by any producer can complete it. □

The new solver therefore improves the work bound and the negative certificate; it does not increase successful inputs merely by replacing that old minimum-fill routine. It also avoids the old producer's potentially large partial fill work once full completion is impossible.

2.3 Exact work families

For the $m \times (m+1)$ chain matrix $a_{i,i} = a_{i,i+1} = 1$, the new pass visits $2m$ source incidences and then at most $2m$ removal incidences. The old producer rebuilds row and column totals at every pivot; its native `entry_scans` counter is $2m(m+1)$ on this family. The same count holds for the rectangular fan with $a_{i,i} = 1$ and $a_{i,m} \geq 2$. These are source-incidence planner bounds; no crossing-count interpretation or end-to-end knot-recognition speedup follows from the matrices alone.

3 Compiling saturated bundles without expanding their genus

A parallelity component with every base boundary circle attached to the core differs fundamentally from one with an unattached boundary interval. The incoming hierarchy-kernel report proves that a proper-boundary attachment adds only relative one-handles. It also shows that a fully attached bordered base requires at most one additional two-handle, but leaves that handle's attaching curve uncompiled [4]. The result of this section supplies the geometric intersection data needed to use that remaining handle in the maintained cancellation model.

The new information is more precise than a relation in a fundamental-group presentation. We identify the unsigned intersections of an *embedded* residual attaching curve with the original core cocores. Its potentially numerous additional one-handles can then be omitted from the cancellation matrix, while their number is retained exactly. The resulting certificate determines an unmarked handlebody census. It does not specify the position of every inherited boundary-pattern curve after the cancellations.

3.1 The geometric input and its source product charts

Let K be a compact orientable three-manifold supplied with an actual handle decomposition containing zero-, one-, and two-handles and no three-handles. Write H for the union of its zero-

and one-handles. Thus H is a disjoint union of handlebodies; we do not initially assert that the components of K are handlebodies. For an original one-handle h, let D_h denote its cocore disc. The original two-handle attaching curves are embedded, pairwise disjoint curves on ∂H.

Suppose that orientable total interval bundles

$$p_i : E_i \longrightarrow B_i$$

are attached to K, where every B_i is a connected compact surface with nonempty boundary. Their complete intersections with the core are $p_i^{-1}(A_i)$, for compact one-dimensional submanifolds $A_i \subseteq \partial B_i$. Distinct bundle interiors are disjoint, and the interfaces have compatible corner collars and no unrecorded intersections. Adjacent source ports have already been amalgamated at their actual common corners, so the connected components of A_i are genuine maximal closed intervals or whole boundary circles.

For an attached component, $A_i \neq \varnothing$. Put

$$a_i = \#\{\text{interval components of } A_i\}, \qquad c_i = \#\{\text{circle components of } A_i\}, \qquad b_i = |\pi_0(\partial B_i)|.$$

We call the component *saturated* if $A_i = \partial B_i$ and set $\varepsilon_i = 1$ in that case; otherwise set $\varepsilon_i = 0$. For a genuine boundary submanifold this is equivalent to $c_i = b_i$, in which case necessarily $a_i = 0$.

Definition 3.1 (Compatibility with the source cocores)**.** The attaching charts are compatible with the original core cocores if, after the fixed compatible rounding and amalgamation of their specified corner contacts, their entire port rectangles and collars lie in the surviving part of ∂H outside the original two-handle attaching regions, and every core-facing port has one of the following product descriptions:

1. A core-zero port lies in the boundary of its specified zero-handle and misses all ∂D_h.
2. A core-one port is a rectangle whose base-arc coordinate runs between the two ends of its specified one-handle and whose other coordinate is the interval-bundle fibre. It meets ∂D_h, for that one-handle h, in one complete fibre, and misses every other $\partial D_{h'}$.

The statements concern the entire rectangles, including their collars, and not only the signed words of their midpoint paths.

These are the original source charts used by the maintained disk-bundle absorption operator [3]. A core-one port comes from the face-sector side of an edge-cap polygon. The transverse coordinate of the source one-handle runs from one tetrahedron chamber to the other; the bundle fibre runs through the gap between parallel normal sheets. A middle transverse cocore therefore cuts the port in one complete fibre. The source tetrahedron–face flags distinguish its two ends even for a self-pairing. This product fact justifies Definition 3.1 for the native adapter.

The midpoint section is preserved by a fibre reversal. Consequently a whole attaching circle has a well-defined embedded midpoint curve γ_{ij}, with actual unsigned incidence

$$n_{ij,h} = |\gamma_{ij} \cap \partial D_h| = \#\{\text{core-one ports on that circle owned by } h\}. \tag{1}$$

The finite source walk also verifies that consecutive port segments join and that the circle closes. Neither an exponent sum nor a freely reduced word can replace the count in (1).

The cocore compatibility is a substantive hypothesis. For an arbitrary annulus parametrization and an arbitrary meridian system, a cocore boundary can have an additional component running parallel to the annulus core. Midpoint intersection numbers alone would then fail to certify that an entire band foot avoids that boundary. We compile the new rows against the original source cocores, before any disc slides, so that the verified rectangle description remains available.

3.2 Relative bands and the contribution of a saturated base

We recall the relative count from the incoming report, including the geometric reason for it. This count is prior work within the project; the residual incidence theorem below is the extension developed here.

Lemma 3.2 (Relative band count)**.** *Let B be a connected compact bordered surface and let $\varnothing \neq A \subseteq \partial B$ have a interval components and c circle components. Set $\varepsilon = \mathbf{1}_{A=\partial B}$. An orientable total interval bundle over B, attached along its vertical bundle over A, admits a relative decomposition with*

$$h_1 = a - \chi(B) + \varepsilon, \qquad h_2 = \varepsilon, \tag{2}$$

and no additional zero- or three-handles. The one-handles connect the $a + c$ initial interface collars. In particular

$$\lambda = h_1 - (a + c) + 1 = 1 - \chi(B) - c + \varepsilon \geq 0.$$

Proof. First suppose that $A \neq \partial B$. Triangulate the pair (B, A) and choose a boundary edge outside A. Root a spanning tree of the dual triangle graph at a triangle incident to that edge. Collapse this triangle across the chosen boundary edge, and subsequently collapse each triangle across its edge to its already removed parent. These are free edges when used. No chosen edge lies in A, so the result is a connected graph containing A, with its relative regular neighbourhood homeomorphic to B.

Begin with collars of the components of A. In a vertex-disc and edge-band neighbourhood of the graph, absorb every additional vertex disc along a forest rooted at those collars. This leaves only surface bands. Since the initial collar has Euler characteristic a and each band decreases Euler characteristic by one, there are $a - \chi(B)$ bands. Their incidence graph connects the $a + c$ initial collars. The interval bundle over each band is a three-dimensional one-handle. The interval bundle over an initial collar is merely an outward collar of the given interface and is absorbed into the core. For a circular component, its loop preserves orientation in the base collar, so orientability of the total space forces trivial fibre reversal around it; its vertical interface is therefore an annulus with a product collar. The interval case is a product directly.

If $A = \partial B$, remove the interior of a small disc Δ in B. The punctured surface B° has a free boundary circle and $\chi(B^\circ) = \chi(B) - 1$. The preceding argument gives $1 - \chi(B)$ one-handles, since now $a = 0$. Restoring $p^{-1}(\Delta) = \Delta \times I$ attaches one two-handle along $\partial\Delta \times I$. A possibly twisted surface band still lifts to a three-dimensional one-handle; the orientability assumption concerns the total space. This also proves the nonorientable-base case. □

For a saturated base of orientable genus g with b boundary circles, $h_1 = 2g + b - 1$. For a nonorientable base with k crosscaps and b boundary circles, $h_1 = k + b - 1$. These can be large numerical quantities. The argument gives an existence statement for the relative bands; it does not require us to write a list with one entry for every band.

3.3 The geometric residual incidence theorem

Theorem 3.3 (Residual incidence from the source annuli)**.** *Assume the geometric input of Section 3.1, including compatibility with the original core cocores. Let $E \to B$ be saturated, with attaching-annulus midpoint curves $\gamma_1, \ldots, \gamma_b$. There is a relative handle presentation with $r = 1 - \chi(B)$ new one-handles and one residual two-handle R such that:*

1. *Its attaching curve α_R has original-core incidence*

$$|\alpha_R \cap \partial D_h| = \sum_{j=1}^{b} |\gamma_j \cap \partial D_h|. \tag{3}$$

2. *It meets the cocore of each of these r new one-handles exactly twice.*
3. *Original core attaching curves and residual curves belonging to other bundle components miss all these r new cocores.*

All the attaching curves occur simultaneously as disjoint embedded curves on the boundary of the enlarged zero/one-handle union.

Proof. Use the punctured surface $B^\circ = B \setminus \operatorname{int} \Delta$ from Lemma 3.2. Its b old boundary circles are the attaching set, and its only free boundary circle is $C = \partial\Delta$. Choose its relative presentation as the regular neighbourhood of the old boundary collars and r bands. The inner edge of each collar remains one of the old boundary circles. Every other boundary segment must therefore lie on C. In particular, *both long sides of every band occur on C, each exactly once.*

We arrange the entire band feet to avoid the old cocores. First extend the old cocore discs through the added interface collars by product strips. By Definition 3.1, their traces on each vertical annulus are finitely many complete interval fibres. On the free edge of the corresponding base collar these fibres give a finite set of marked points.

Shrink and relocate the finitely many band-foot intervals into small unmarked arcs, preserving their cyclic order on that free edge. An orientation-preserving circle homeomorphism realizes this relocation. Interpolate it across the collar to the identity on the *actual attaching edge* of B. Thus the resulting collar isotopy fixes A pointwise. Its lift fixes the given attaching annulus pointwise and does not modify the external gluing map. Since the cocore traces are entire fibres, the lifted foot rectangles and sufficiently small neighbourhoods of them miss all old cocore boundaries. Avoiding just the midpoint points without this product hypothesis would not suffice.

The free circle C follows the outer edge of each old collar except where a band foot has replaced a short subarc. Every old cocore crossing lies outside those feet and is encountered exactly once. The portions on the bands lie in new one-handles and add no traversal of an original core one-handle. This proves (3).

Lift the bands to one-handles and restore $\Delta \times I$. The midpoint section of its attaching annulus gives the embedded curve α_R. Each long side of a band crosses the lifted band cocore once, so the two sides give exactly two geometric intersections. Twisting a surface band can change the signs of these passages but cannot change their unsigned number.

The original attaching curves lie outside the interface collars, and different bundles have disjoint charts. Their residual curves can therefore be constructed disjointly, and no such curve uses another bundle's bands. The entire construction adds one-handles before two-handles and creates no three-handle. It is consequently a genuine geometric handle presentation with all the asserted intersections. □

For B a disc, $r = 0$ and (3) is exactly the earlier disk-absorption formula [3]. For a higher-genus or nonorientable base, the theorem compiles the same finite kind of row even though the residual curve may run through a very large family of new handles. It neither produces a short freely reduced word nor needs one.

3.4 Connecting different cores and counting the private handles

Contract a spanning forest of the original zero/one-handle graph. Let its connected components be indexed by j. Component j has v_j original zero-handles, e_j original one-handles, and f_j original two-handles. Its remaining original one-handle count is

$$d_j = e_j - v_j + 1.$$

Contracting a tree handle deletes its incidence column and retains the other original cocores. It does not freely reduce any attaching curve.

Form the bipartite attachment multigraph Γ: its vertices are the original core components and the attached bases, and each connected interval or circle interface gives one edge. Repeated attachments are retained as separate edges. For a connected component C of this graph, let q_C be its number of core vertices and put

$$H_C = \sum_{B_i \in C} (a_i - \chi(B_i) + \varepsilon_i), \qquad p_C = H_C - q_C + 1. \tag{4}$$

Lemma 3.4 (Connector forest)**.** *The enlarged zero/one-handle union indexed by C is connected. Exactly $q_C - 1$ of its added one-handles can be contracted as a spanning tree joining the original core components. Its remaining added one-handles number $p_C \geq 0$.*

Proof. The relative-band graph for one attached base connects all its interface collars. After those collars are absorbed into the core, it therefore connects precisely the core components incident to that base. Doing this for all bases gives the same connected-component relation as Γ. Higher-index handles do not join components. Hence the added one-handle graph on the q_C original core pieces is connected. Choose a spanning tree, which has $q_C - 1$ edges, and contract its corresponding handles. The remaining number is $H_C - (q_C - 1)$, as claimed. □

We call these remaining added handles *private* to distinguish them from the original core columns on which the cancellation search will operate. Initially each private column has either only zero entries or a single entry equal to two in its own saturated residual row. Tree contraction merely deletes some added columns and leaves this property intact.

The correction by $q_C - 1$ is essential. A single saturated annulus has $1 - \chi(B) = 1$ relative band. If its two boundary annuli attach to distinct core components, that band joins them and is contracted; there is no surviving private handle. More generally, one saturated base meeting s distinct cores leaves $1 - \chi(B) - (s - 1)$ added handles after those mergers. This is $2g + b - s$ for an orientable base and $k + b - s$ for a nonorientable base.

3.5 The finite active matrix and the handlebody census

The full geometric positive-pivot operation at an entry $n_{rg} = 1$ has unsigned update rule

$$n'_{ih} = n_{ih} + n_{ig} n_{rh} \qquad (i \neq r,\ h \neq g). \tag{5}$$

The maintained geometric disc-slide theorem realizes this rule by embedded disc slides followed by a one/two-handle cancellation [3]. Its hypothesis is the actual one-point intersection, not a unit algebraic intersection number.

An initially even column stays even under (5). More generally, an initial column with entries only zero or at least two can never acquire a unit entry; see Lemma 2.5. Thus no private column can ever be used as a pivot. The original-column submatrix evolves autonomously under the same rule, since an update on an original column refers only to entries in that column and in the selected original pivot column.

Proposition 3.5 (Active incidence compilation)**.** *After the forest contractions above, form a matrix N as follows. Keep the rows of the original core two-handles, restricted to the surviving original one-handles. Append one row for every saturated base, with entries given by (3). Do not append any row for an unsaturated base and do not represent any private column.*

The full relative geometric presentation admits a positive-pivot schedule eliminating every two-handle if and only if N does. Any successful schedule for N certifies actual geometric cancellations in the full presentation. This equivalence remains true when all positive entries are capped at two.

Proof. Private columns cannot pivot, and deleting them leaves the updates on the other columns unchanged. A full schedule therefore restricts to a schedule on N. Conversely, execute a schedule for N on the full matrix. Its selected entries are still one, so the geometric disc-slide and cancellation theorem applies at every step. All rows are removed. The map $n \mapsto \min(2, n)$ preserves addition and multiplication in the saturated nonnegative semiring, and detects the predicate $n = 1$ exactly. It therefore preserves the argument. □

Theorem 2.2 supplies the complete finite procedure for full-row elimination in this raw model. In particular, the large matrix implicit in the relative presentation need never be materialized: the producer and checker work with the original core columns, the original rows, and one additional finite row per saturated base.

Theorem 3.6 (Certified whole-cut handlebody census)**.** *Assume the source geometric contract, no core three-handles, and bordered bases. Suppose the compiled active matrix has a verified schedule eliminating all its rows. Then each component X_C indexed by the attachment graph is a handlebody, of genus*

$$G_C = \sum_{j \in C} d_j + p_C - \sum_{j \in C} f_j - \sum_{i \in C} \varepsilon_i \tag{6}$$

$$= \sum_{j \in C} (e_j - v_j + 1 - f_j) + \sum_{i \in C} (a_i - \chi(B_i)) - q_C + 1. \tag{7}$$

Every unported bordered bundle is a separate handlebody of genus $1 - \chi(B_i)$ and contributes its entire authenticated multiplicity. These data determine the complete genus histogram of the cut.

Proof. After contracting the two forests, the connected zero/one-handle union for C is a handlebody with $\sum_j d_j + p_C$ one-handles. There are $\sum_j f_j + \sum_i \varepsilon_i$ two-handles. By Proposition 3.5, the checked schedule cancels all of them geometrically, leaving exactly the number of one-handles in (6). Substitution of (4) gives (7).

An orientable total interval bundle over a connected bordered surface is a handlebody: choose proper arcs cutting the base to a disc and lift them to cocore discs. There are $1 - \chi(B_i)$ arcs, including for a nonorientable base. An unported component has no intersection with the core or with another bundle component, so it must be counted separately with its full multiplicity. □

There is an independent Euler check. The intersection of the core with a proper interval interface is a disc, and with a complete circle interface is an annulus. Hence

$$\chi(X_C) = \sum_{j \in C} (v_j - e_j + f_j) + \sum_{i \in C} \chi(B_i) - \sum_{i \in C} a_i,$$

which equals $1 - G_C$ by (7). This equality checks the count; it does not substitute for the cancellation proof. In particular, the conditional expression $G_C = 0$ is not a ball certificate when the active matrix has a surviving row.

3.6 Examples and boundaries of the conclusion

A saturated annulus joining two solid tori. Let its boundary annuli attach to solid-torus cores whose standard meridians meet their midpoint curves p and q times respectively. Choose the usual embedded slopes with these positive geometric winding numbers. The connecting band merges the two cores and is contracted. The active matrix is the single row (p, q), with no private columns. If $p = 1$ or $q = 1$, one geometric cancellation proves that the union is a solid torus. The genus computation is $2 + 0 - 1 = 1$.

For $p = 2$ and $q = 3$ there is no unit pivot. This refusal is necessary for this example: van Kampen gives

$$\langle x, y \mid x^2 = y^3 \rangle,$$

which maps onto the nonabelian group S_3 by sending x to a transposition and y to a three-cycle. Its abelianization is $\mathbb{Z}$, so first homology alone would not distinguish it from the positive solid-torus cases.

A saturated Möbius bundle. The orientable interval bundle over a Möbius band is a solid torus, with the vertical boundary representing the square of its core generator. Attach that annulus to a solid-torus core along an annulus with one geometric meridian intersection. The relative description has one private one-handle and one residual row whose active part is (1). Cancelling against the original core handle leaves one private handle, proving a solid torus. The corresponding relation can be written $x = a^2$, but the positive conclusion comes from the embedded cancellation.

If instead the same vertical annulus is attached to a nullhomotopic annulus on the boundary of a ball, the active matrix has one *empty* row. Its private column has incidence two. No cancellation is authorized. Indeed the resulting group is $\langle a \mid a^2 \rangle \cong \mathbb{Z}/2$, whereas its Euler characteristic is one and its boundary is a sphere. This is a concrete reason to keep the surviving-row obligation separate from the conditional genus arithmetic.

Arbitrarily large binary genus. Let B be orientable of genus g with one boundary component, and attach its vertical boundary annulus to a solid-torus core along a curve meeting a standard meridian once. There are $2g$ private handles and one active unit entry. The residual cancels the original core handle, leaving a handlebody of genus $2g$. For $g = 2^L$ the genus counter requires $O(L)$ bits, while the active matrix still contains just one entry. This is a family of explicitly specified relative topological attachments. It does not assert that every member has a normal-cut source triangulation with $O(L)$ tetrahedra.

What successful augmentation does and does not imply. No core handlebody conclusion is assumed by the compiler, but a successful full column-peeling schedule restricts to one for any subset of its rows: deleting the other rows can only make its pivot columns less occupied. Thus successful joint cancellation implies raw cancellability of the original core rows themselves. Adding saturated residual rows cannot rescue a core outside that raw cancellation class. This observation also justifies an early filter: an obstruction in the original core matrix permits this consumer to stop before requesting the ordered-boundary queries. It is a refusal by this consumer, not a proof of non-handlebodyness.

If the active matrix stalls, disc slides that reduce geometric intersection numbers, another meridian system, or a different handle presentation might still succeed. Closed bases and core three-handles remain outside the present operator. A successful histogram concerns the *entire supplied cut*, including isolated components. A knot verdict additionally requires the appropriate cutting-surface and peripheral information together with authenticated provenance from the input knot diagram.

Finally, the private bands and the residual curve are constructed by existence in the proof, while their required incidences are compiled from source charts. The certificate does not provide a source address for an arbitrary private cocore, nor the transport of a later boundary pattern through the chosen band decomposition. Closing those geometric interfaces, controlling surface discovery, and bounding all hierarchy stages remain necessary to reach the quasi-polynomial objective discussed in Lackenby's program [10].

3.7 A conditional extension for observer curves

The private-handle count can be supplemented by certain exact incidence data when those data are already available. The following algebraic observation identifies a limited interface for future curve transport; it does not provide the missing initial geometric data.

Lemma 3.7 (Observer families under column peeling)**.** *Let A be the nonnegative integer incidence matrix of m task rows, and let Y be a nonnegative integer matrix of observer rows on the same columns. Suppose the private columns are partitioned into families F_ℓ such that all columns in one family are identical across both A and Y initially. Assume every private task entry is zero or at least two; observer rows are never eligible for pivots.*

Under any positive-pivot schedule on active columns, equality within each private family is preserved. One representative column and the binary multiplicity $|F_\ell|$ therefore suffice for that family. If the schedule uses task column leaves, let $W \geq 2$ bound every exact *initial entry of A and Y. After $t \leq m$ pivots, each surviving observer entry is at most*

$$W(1+W)^t,$$

and hence has $O((m+1)\log(W+1))$ bits, independently of the private-family multiplicities.

Proof. For a pivot (r, g) the observer update is

$$Y'_{u,h} = Y_{u,h} + Y_{u,g}A_{r,h} \qquad (h \neq g). \tag{8}$$

The task rows follow the same positive update rule. Equal columns receive equal additions in every surviving row, so family equality is invariant. Private task columns cannot acquire a unit entry, by Lemma 2.5, and their multiplicities are unchanged.

For a task column leaf, $A_{r,g} = 1$ and $A_{i,g} = 0$ for every other live task row i. Thus task rows never fill: every surviving task entry is an initial entry and is at most W. If M_t bounds the observer entries after t pivots, (8) gives $M_{t+1} \leq (1+W)M_t$. Starting with $M_0 \leq W$ proves the bound. No update sums over the columns of a private family, so its multiplicity contributes no factor to this recurrence. □

Geometric interpretation. Suppose the observer rows are authenticated unsigned cocore incidences of embedded curves disjoint from all task attaching curves. Choose the attaching collars sufficiently small to avoid the observers. The geometric positive-pivot proof can likewise choose its sliding bands in a sufficiently small neighbourhood of the donor task curve, avoiding the finite union of observer curves. Each disc slide then adds one copy of the relevant cocore's observer intersections, giving exactly (8). The observers survive the attachment and cancellation because they avoid the attaching region. This applies the same embedded disc-slide argument as the task-row formula [3].

What additional input is required. The update must use *uncapped* donor entries: a value recording only "at least two" suffices for pivot admissibility but not for exact observer counts. The number of families and their initial observer coordinates must also be supplied and geometrically authenticated. Scalar genus and multiplicity profiles do not supply them or bound the number of required families. Multiplicity itself still requires $\log(|F_\ell| + 1)$ bits of storage. Even when the hypotheses hold, the output consists of cocore intersection counts; it does not determine cyclic order, source addresses, or the embedded position of an arbitrary boundary pattern. The current implementation does not produce these initial observer data or implement this conditional extension.

4 Encoded cost and the remaining global complexity gap

The size parameters must distinguish four different objects: the knot diagram, an explicit triangulation at one stage, a binary normal vector, and a supplied proof. Let n be the crossing count of a diagram, t the number of tetrahedra in a particular source triangulation, and

$$b = \max\bigl(1, \max_{i,j} \lceil \log_2(1 + x_{ij}) \rceil\bigr)$$

the maximum coordinate bit length. The explicit coordinate input has $7t$ entries, so reading it already costs $\Omega(tb)$ bits in the worst case. Neither t nor b is silently replaced by n during a hierarchy.

For the finite relative compiler, let s count the explicit core handles, source attaching-cycle occurrences, ports, base-profile records, and ordered attachment letters. Let $B \geq \log_2(s+2)$ bound the bit lengths of all source labels, profile integers, and resulting counters, and let $M(B)$ denote a multiplication cost for B-bit integers. The number of private one-handles is a *value* stored in one of these integers; it is not included in s as a list of handle objects.

4.1 What the finite compiler actually bounds

Proposition 4.1 (Finite compressed compilation cost)**.** *Given authenticated finite source geometry, base profiles, and an ordered attachment skeleton of size (s, B) satisfying the hypotheses of Theorem 3.3, the active incidence matrix has at most one additional row per attached base and no additional column. It can be constructed and independently replayed in time polynomial in s and B, independently of the numerical private-handle count and of the numerical multiplicity of an isolated bundle.*

For the delivered straightforward implementation, a conservative bound is $O(s^2 M(B))$ bit work under bounded-cost dictionary indexing of the explicit labels, with $O(s)$ stored finite records and their B-bit payloads. Deterministic sorting or balanced dictionaries preserve a polynomial bound. The square factor is an implementation bound for repeated component scans, not an inherent lower bound.

Proof. An original core attaching cycle is scanned once to form its row after deleting forest columns. Every source port occurs once in the ordered attachment components. A saturated row is the capped sum of the core-one ports in its own circles. Since the profiles partition ports, these scans cannot create a list proportional to the base genus. The active nonzero count is bounded by the number of original cycle occurrences plus the number of core-one ports.

Union–find on the explicit core and attachment records determines which original cores are joined. The delivered code then scans profiles, attachments, handles, and rows for each joined component. There are at most s components and s records per scan. The genus and Euler formulas use sums and products of encoded integers; charging each such operation by $M(B)$ gives the stated conservative bound. The checker reconstructs the same mathematical quantities with separate code and the same type of finite scans. Isolated multiplicity contributes a multiplication to the histogram rather than repeated graph vertices. At no point is a private handle enumerated. □

In the maintained normal-cut source model, core handles and total source incidences are $O(t)$, and the source port bound is $66t$ [3]. In particular the active matrix has at most $12t + 66t$ rows and $8t$ original columns, using conservative handle and ported-base bounds. These are upper bounds, not typical dimensions. The inherited orbit computations and their encoded output sizes remain separate costs. Agol–Hass–Thurston orbit counting is the foundational compressed connectivity tool[12]; calling a finite adapter polynomial does not supply a new proof for every upstream discovery path.

For the matrix layer itself, the stronger bound from Section 2 is available. With m rows, k active columns, and I source nonzeros, discovery takes $O(m + k + I)$ expected dictionary operations, apart from final label sorting. The counts used for leaf detection have $O(\log(m+1))$ bits after individual capping. A literal count of 2^{16384} costs time to read, but never causes that many updates.

The full census producer has the form

$$D_{\text{source}}(t, b) + D_{\text{boundary}}(t, b) + \text{poly}(s, B),$$

where the first two terms are actual upstream discovery costs. Its replay has the analogous source-verification terms plus the cost to read the supplied orbit traces. A trace-operation cap bounds the number of trace records admitted; it is not a hard bound on time, bytes, or malformed input nesting. The optional precheck can omit the boundary term on a certified original-core obstruction. Successful cases still pay for the complete source and attachment information.

4.2 Existence of a large handle system versus outputting it

Theorem 3.6 certifies an invariant of a genuine finite manifold without printing every disc in a complete disc system. That distinction is unavoidable when a profile permits 2^B private handles. A certificate consisting of their count and the residual-incidence theorem can be short. A literal output with one named cocore per handle has an immediate $\Omega(2^B)$ output-size lower bound.

This is why the present compiler cannot simply be declared a general hierarchy-transition routine. A later surface might need to cross, or refer to, individual private cocores. A short *addressing and transport scheme* for those cocores is then an additional obligation. The current proof neither allocates those objects nor gives source addresses for all of them. The arbitrary-genus example proves that the terminal census can remain compressed; it does not bound the cost of all possible later geometric queries.

4.3 A conditional composition theorem

The following elementary theorem makes the global target falsifiable. Its hypotheses are the missing interface bounds, rather than conclusions drawn from the local experiment.

Theorem 4.2 (Sufficient bounds for a quasi-polynomial composition)**.** *Consider a correct recognition algorithm on n-crossing diagrams with at most*

$$H(n) = L(n)(g(n) + 1)^{L(n)}$$

charged control steps. Suppose every step tries at most $C(n)$ candidates, each candidate and its certificate have encoded size at most $S(n)$, and discovery, transitions, and replay together take at most a fixed polynomial in $n + C(n) + S(n)$ per charged step. If

$$C(n), S(n), g(n) \le n^{O(1)}, \qquad L(n) = O((\log n)^a),$$

for a fixed $a \ge 1$, then its total time is

$$2^{O((\log n)^{a+1})}.$$

In particular $a = 1$ suffices for the February slide exponent, and $a = 2$ suffices for the March slide exponent.

Proof. Take logarithms of the step bound:

$$\log H = \log L + L \log(g+1) = O((\log n)^{a+1}).$$

Multiplying by a fixed polynomial in the stipulated polynomial quantities adds only $O(\log n)$ to the logarithm of the running time. A polynomial preprocessing or final verification phase is absorbed in the same bound. □

The hypothesis includes *finding* the candidate surfaces and compression discs. A small verified certificate proves efficient checking of that certificate; it does not bound a search that might inspect exponentially many alternatives. Similarly, a polynomial formula in t and b does not help globally if successive stages make either quantity exponential in the original n.

Even granting $g(n) = n^{O(1)}$, the weaker bound $L(n) = O(n)$ gives $2^{O(n \log n)}$ by this calculation. Reducing a local quadratic pass to a linear pass changes the polynomial factor. It does not change this reset exponent. To improve that exponent, one must either shorten the hierarchy, reduce the number of resets, or amortize several charged steps by a stronger theorem. The ambient-block result later in this article explores the third possibility under explicit extra geometric hypotheses.

4.4 Why simple potential shortcuts do not close the gap

Replacing a lexicographic vector by its sum requires a proof that a repair does not recreate large later entries. The sum has no monotonicity merely because the first altered coordinate decreases. Likewise, replacing an integer intersection count I by $\log(I+1)$ does not bound a process that can delete only one intersection at a time: the logarithmic decrement can be arbitrarily small. The potential theorem below retains an honest numerical intersection term and gives a concrete binary-size obstruction.

These observations guide implementation priorities. Fast local replay is valuable and directly measurable. The global objective additionally needs a geometrically justified amortization argument, with bounds expressed in the original input size and maintained throughout every transformation.

5 Implementation and independently replayed certificates

The delivered extension is opt-in and uses the Python standard library. The pinned native runtime is included for reproduction. Regina is used only by the explicit-cut oracle, and is not imported by the new producer or verifier. The default recognizer and its schedules are unchanged.

5.1 The producer

`normal_cut_saturation` accepts the same source triangulation and normal-coordinate contract as the existing normal-cut parallelity operator. Its logical sequence is:

1. Construct the source parallelity decomposition, with four orbit queries and optional upstream certificates. Reject unsupported core three-handles or closed bundle bases as inconclusive.

2. If the caller enables the original-core precheck, form its fixed forest and capped source incidence rows. A nonempty stopping core permits an early inconclusive result; no ordered boundary information is needed.

3. Obtain the ordered attachment skeleton, requiring at most two more orbit queries. Source-zero-port cases may require none.
4. Compile all original core rows and all saturated residual rows jointly against the original source cocores. Record the exact private one-handle count in each connected attachment component.
5. Run unit-column peeling. Complete elimination authorizes the handlebody histogram and Euler check. A nonempty residual authorizes only an inconclusive result for topology.

The discovery cycle allowance is shared across the upstream queries. The optional precheck defaults to false, so the principal audit and certificate format are directly comparable to the complete six-query path. Its successful path returns the same whole-cut certificate as the ordinary path; its early obstruction uses a smaller four-query schema.

5.2 What is and is not trusted

The whole-cut certificate has four proof layers and an exact conclusion:

Layer	Claimed data	Independent check
Source parallelity	Core handles, bundle bases, fibre cover and port ownership	Reconstruct the source geometry and verify the four orbit proofs
Ordered boundary	Maximal interval/circle interfaces and signed source port walks	Independently reconstruct and replay the endpoint and gap data
Relative compiler	Forest, residual rows, attachments, component mergers, private counts	Rebuild finite source paths and relative counts without calling the compiler
Peeling	Row/column deletion schedule and final residual	Timestamp scan of immutable source incidences; stopping inequalities
Conclusion	Exact genus histogram or exact raw-obstruction statement	Recompute ranks, multiplicities, Euler characteristic, and result fields

The enclosing verifier imports neither the relative compiler nor the peeling producer. The existing source verifiers reconstruct the source geometry independently rather than accepting a geometry dictionary because it carries a matching checksum. Both source proofs are bound to the *same* input triangulation and normal vector. The finite compiler verifier alone is conditional on its supplied geometric inputs; it is the enclosing verifier that closes this source-authentication boundary.

A source fingerprint guards against accidental substitution, but it is not the mathematical reason a conclusion is correct. The reason is the reconstruction, the geometric theorems, and the exact incidence replay. The certificate includes no unchecked Boolean saying that a residual row is geometrically valid or that a component is a ball.

The early certificate follows the same principle with fewer layers. It contains the authenticated source parallelity proof, a finite core forest and row record, a peeling certificate with nonempty residual, and the exact conclusion `INCONCLUSIVE`. Its checker independently reconstructs the core rows before verifying the stopping set. By Corollary 2.4, no eventual saturated row can make that fixed core subset peel.

5.3 Status and integer semantics

`READY` in the finite compiler means that a relative presentation has been compiled. It is not a topology result. `COMPLETE` in the whole-cut consumer means that every row has cancelled and that the exact handlebody census follows. `OBSTRUCTED` in the matrix layer means that full-row raw positive cancellation is impossible. At the topology boundary it becomes `INCONCLUSIVE`.

The source matrices retain empty rows. Omitting an empty residual row would incorrectly turn the Möbius-on-a-ball example into a ball certificate. Counts are capped individually at two only for incidence logic. The total of those capped counts remains uncapped for deletion bookkeeping; saturating the total itself would lose the information needed to know when a column becomes a leaf.

Encoded integers use the existing native hexadecimal convention for very large values. The code never disables Python's global decimal conversion limit. Genus, multiplicity, and Euler arithmetic use exact integers. Boolean values are rejected where exact integers are required. Certificate record keys and canonical lists are checked, and caller cancellation exceptions propagate rather than being swallowed as false proofs.

5.4 Why integration remains opt-in

Replacing the old minimum-fill routine globally would change the meaning of its partial output. On an obstructed matrix, the old routine can make many legal partial cancellations, whereas column peeling may stop immediately with a canonical residual. Neither method promises maximum partial cancellation. A consumer using only the full-row verdict can switch safely after the stated theorem and tests; a consumer using the old partial schedule needs a separate interface review.

The supplied integration patch therefore adds new modules without changing the default solver or its old proof schemas. A caller can explicitly invoke the extended whole-cut consumer and independently replay its result. Promoting it into a knot-recognition schedule would additionally require the selected-surface and peripheral correctness obligations of that schedule. The integration notes list the exact files and test commands.

6 Validation and measured performance

The mathematical statements above do not depend on experiment. The experiments check that the code implements their hypotheses and conclusions, that the new source adapter agrees with actual cut manifolds, and that a claimed kernel improvement survives complete-call measurement. The preserved files contain every measured sample, including unfavorable comparisons and inconclusive cases.

6.1 Baseline and algebraic validation

The pinned repository's full native suite passed all 1,663 tests in an isolated checkout with its original surrounding directory structure. Its log also reports 13,846 expanded-sheet topology comparisons. This is a baseline check, not a claim that the new modules were already integrated into that suite. An initial run from an incomplete copied directory could not resolve tests' repository-relative fixtures; it was discarded as an invalid test setup and the full checkout run was used instead.

The column-peeling validation includes the following distinct checks.

- 31,106 small nonnegative matrices were compared with an independent recursive search over every allowed positive-pivot order.
- 600 rectangular cases were checked by separately enumerating permanent-one full-row minors.
- 512 binary matrices were checked against the union of all stopping subsets and against source-order reversal.

- 24,508 matrices were compared with the pinned minimum-fill planner. All full-row verdicts agreed, and every successful baseline run used only zero-fill pivots, as the completeness proposition predicts.

These populations overlap and must not be added as if they were disjoint independent inputs. Further tests exercise empty rows, empty matrices, large sparse labels, 4,096-bit counts, inert columns, source immutability, forged certificates, strict types, and propagated cancellation exceptions.

The relative-bundle tests include an actual layered-solid-torus normal cut with independent source replay, and separate standard relative topological models: a saturated disc, Möbius band, once-punctured torus, annuli joining two and three core components, and a genus 2^{4096} profile. The latter is a conditional geometric-profile test, not an assertion that the native triangulation realizes that enormous base with a constant number of tetrahedra. Eleven hostile relative certificate mutations are rejected. The enclosing-wrapper tests independently exercise source substitution, forged conclusions, zero-cycle discovery budgets, exact and one-short replay budgets, hexadecimal transport, and replay with producers disabled. The combined extension suite passes all 37 test methods. Its seven precheck methods include the complete enabled/disabled corpus comparison; the same cases are not counted as additional independent topology inputs.

6.2 The fresh 106-case native audit

Outcome	Pinned consumer	Saturated compiler
Complete handlebody census	78	85
Unresolved annular interface	11	0
Raw cancellation failure/obstruction	7	11
Unsupported core three-handles	10	10
Total	106	106

Table 3: The same supplied normal cuts, freshly run against both consumers. The four newly classified raw obstructions are former annular refusals; they are not negative topology declarations.

All 78 prior complete censuses are preserved. Of the 11 old annular refusals, seven become complete and four now receive precise raw obstructions. Every one of the 96 compiled whole-cut certificates replays independently, including the 11 obstruction conclusions. The ten three-handle cases return before compilation.

Let H_g denote an orientable handlebody of genus g. The newly complete cases are listed individually, so the aggregate gain can be inspected.

Native case identifier	Exact cut census
`pilot/fibonacci_lst_05/0`	$H_1 \sqcup H_2$
`pilot/fibonacci_lst_07/0`	$H_1 \sqcup H_2$
`pilot/fibonacci_lst_09/0`	$H_1 \sqcup H_2$
`pilot/lst_4_7/0`	$H_1 \sqcup H_2$
`pilot/lst_7_10/0`	$H_1 \sqcup H_2$
`fibonacci_lst_09/mixed`	$H_0 \sqcup H_3 \sqcup H_3$
`fibonacci_lst_10/mixed`	$H_0 \sqcup H_2 \sqcup H_2$

Table 4: All seven new successes have saved source-bound certificates and agree with independently constructed literal cuts.

The oracle constructs each normal surface in Regina 7.4, calls `cutAlong()`, separates the actual resulting triangulation into components, and calls `recogniseHandlebody()` on each component. The latter returns its genus or -1 for a non-handlebody[13]. The oracle does not use the new relative-bundle presentation. A cap of 4,096 normal discs prevents explicit expansion of the binary families; each permitted oracle call also has a 15-second timeout. All 99 permitted oracle calls completed, with no timeout. All 78 positive results within the expansion cap agree with the oracle. The remaining seven positive binary inputs are independently replayed native certificates, not explicit Regina comparisons. The largest source coordinate has 16,385 bits.

Of the 21 unresolved cuts, the oracle identifies 15 as unions of handlebodies: eight raw obstructions and seven three-handle cases. The remaining six have at least one non-handlebody component. We retain all of these inputs. The 15 unresolved positives show directly that the extended topological method is incomplete, while the mixture of positive and negative topology among raw failures demonstrates why the refusal must remain inconclusive.

6.3 Exact incidence-work improvement

512-row family	Old scans	New source visits	New deletion visits
Rectangular chain	525,312	1,024	1,024
Rectangular fan	525,312	1,024	1,024
Cyclic stopping core	525,310	1,024	0

Table 5: Native work counters for fixed matrices. On the cyclic stopping core the old routine additionally makes 511 partial fill updates; the new routine makes none. The comparison concerns incidence-processing work, not CPU instructions or knot crossings.

For the chain and fan, the baseline rebuilds incidence totals throughout the elimination and records $2m(m+1)$ entry scans. The new procedure makes $2m$ source visits and $2m$ deletion visits. Thus this explicit family exhibits a quadratic-to-linear change in planner incidence work. On the cyclic stopping core the negative certificate is available before any partial fill. No claim is made that a maximum partial cancellation has been computed.

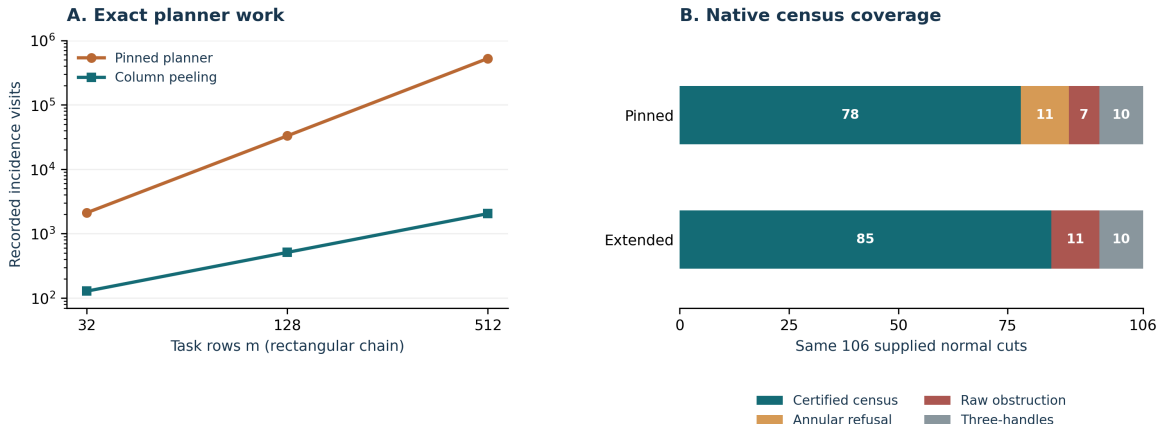


Figure 1: Left: exact recorded incidence work on rectangular chains, including both source and deletion visits for the new method. Right: complete classification of the same 106 native source cases by each consumer. Numerical speedups on the left and topological coverage on the right measure different improvements.

The paired kernel timings used seven alternating-order rounds, with separate A/A repetitions of the unchanged baseline and independent replay timed separately. They ran after the baseline

regression process had exited. The following 128-row samples had comparatively stable controls.

128-row family	Old ms	New ms	Old/new	A/A
Rectangular chain	8.424	0.293	28.73	1.004
Rectangular fan	7.477	0.269	27.78	1.001
Cyclic stopping core	5.075	0.102	49.73	1.024

Table 6: Medians for the planner kernel on Python 3.12.14. Source-matrix construction and JSON output are untimed. Full raw samples and the separate verifier times are preserved.

The 512-row timing ratios are also retained, but their A/A controls range from 1.070 to 1.194. That variation is why exact work counts, rather than the largest wall-time ratios, carry the asymptotic conclusion.

6.4 Complete native calls and the optional precheck

The whole-cut timings include source reconstruction. In the verified task, each repetition creates a fresh certificate and independently replays it. Module import, corpus-file loading, and JSON output are excluded. All calls ran sequentially after the audits had finished; seven alternating rounds and A/A controls were used. The controls measure variability in this shared execution environment, rather than claiming dedicated hardware.

Verified common-success task	Old ms	New ms	Old/new	A/A
`pilot/fibonacci_lst_01/0`	1.952	1.866	1.046	0.943
`fibonacci_lst_05/mixed`	107.069	104.351	1.026	0.960
`relabel/4/0`	432.042	444.604	0.972	1.003
`binary/1/500`	7.438	7.726	0.963	1.011

Table 7: The old and new consumers perform the same complete census task. The final two rows are regressions in this sample. Source discovery and source replay dominate these small active matrices.

These common-success measurements do not support a broad end-to-end speedup claim. The verified ratios range from 0.963 to 1.046, and several controls show comparable variability. The gain in positive coverage is the new geometric construction, while the large kernel ratios apply to growing incidence families rather than every native input.

The optional precheck gives a different, direct whole-call improvement. On all 106 inputs it preserves the 85 successful certificates exactly. It produces six early original-core obstruction proofs. Five inputs actually avoid two boundary orbit queries each, for ten avoided queries in total. The sixth, `pilot/finite_trefoil/0`, has no ports, so the original boundary stage already made zero orbit queries; only later finite work is skipped there.

Verified obstruction task	Full path ms	Precheck ms	Ratio	A/A
`pilot/fibonacci_lst_04/0`	49.713	21.953	2.265	0.910
`pilot/fibonacci_lst_10/0`	97.903	51.890	1.887	1.002
`pilot/lst_4_9/0`	32.936	16.124	2.043	0.989
`pilot/finite_trefoil/0`	5.137	2.866	1.792	0.981

Table 8: Equivalent refusal of this raw certificate method. The early proof establishes an original-core stopping subset and avoids compiling the full residual matrix; it does not claim to output that matrix's canonical residual. The first A/A control is noticeably less stable.

The stopping-set theorem justifies the skipped work. Thus the two-query savings are exact and the roughly twofold verified-call reductions in these samples have an identified cause. They remain measurements of supplied-cut tasks, not of a complete unknot recognizer or a distribution of arbitrary knot diagrams.

7 Retaining ambient surfaces through ordinary compression

The saturated-bundle operator concerns the topology of a supplied cut. A different obstruction to a fast hierarchy algorithm is repeated replacement of later surfaces after an earlier surface is compressed. The hierarchy algorithm of [10, Sections 6 and 9] uses precisely such a replacement step. This section gives an additive repair bound for a restricted, explicitly stated situation: the cut-open surfaces are the pieces of a certified family of surfaces in one fixed ambient manifold. The result does not construct the required families or provide a new unknot-recognition algorithm. Its purpose is to identify geometric data under which the choice of every later ambient surface can be retained.

7.1 The additional gluing obligation

An ordinary sequential hierarchy does not automatically supply simultaneous ambient surfaces. After an earlier cut, a later surface can have unrelated boundary traces on the two copies of the cutting surface. Changing that earlier surface can expose attachments on its opposite side which were absent from the side on which a compression disk was found. A list of homology vectors does not record this information.

We use a *surface atlas* to mean a concrete PL cut-and-reglue presentation, with the surface pieces labeled and cooriented, and with the following additional conditions. Traces bearing the same label agree under every specified face identification. Every internal boundary arc is paired exactly once; unpaired boundary lies on the true boundary of the ambient manifold. Coorientations agree across the identifications. Distinct patches of a single label have disjoint interiors. At the identified edges and corners, the link of each label is a single embedded circle at an interior surface point, or a single properly embedded interval at a boundary surface point, in the corresponding ambient link.

These are conditions on actual charts and gluing maps. In an explicit finite refinement they are finite incidence and link tests. No verifier for an arbitrary binary-compressed surface atlas is implemented here.

In the following, a *cut-open face* is a connected part of a labeled surface cut along its intersections with the other surfaces, realized on one boundary side of a cut piece. It can have positive genus or several boundary components; it is not required to be a polygon. Its interior maps injectively into the corresponding ambient surface. A *seam* is an actual trace of one of the other cutting surfaces. Auxiliary subdivision edges in the atlas are not seams. Thus essentiality below is tested in the whole cut-open face, even when it is authenticated using many polygon charts.

Lemma 7.1 (What an authenticated atlas supplies)**.** *Under the preceding conditions, the quotient of the patches with any one label is a properly embedded cooriented PL surface. In an orientable ambient manifold it is oriented. Suppose an embedded disk in one cut piece has interior disjoint from its boundary faces, and has boundary in the interior of a single cut-open face of one label, disjoint from all seams and other labels. Its image under regluing is an embedded disk whose interior avoids the whole surface family and whose boundary meets only that labeled surface.*

Proof. At a patch interior the required local surface chart is already supplied. Across an

identified face, the paired traces give two compatible local halves of a surface. The edge and corner link conditions supply the local product charts at the remaining points; in particular they exclude branching, self-intersection, and an unpaired internal boundary. The boundary condition identifies the resulting surface boundary with a subset of the true manifold boundary. The coorientations glue, and an ambient orientation converts a coorientation into a surface orientation.

Regluing is injective on the interior of each cut piece. On the boundary of the stated disk, it is injective because this boundary lies inside one cut-open face whose interior embeds in its labeled ambient surface. No interior point of the disk can be identified with a boundary point: the former is off the surface family and the latter lies on it. The disk image is therefore embedded. The same conditions show that it meets no other labeled surface and meets its own label only along its boundary. □

The local link tests in Lemma 7.1 must not be omitted. Agreement of formal boundary chains alone would allow a branched or immersed quotient. Conversely, once the ambient surfaces have been authenticated, the repair theorem below is purely geometric and does not depend on a particular triangulation of them.

7.2 A potential for one fixed ambient family

Let M be a compact orientable irreducible 3-manifold, and let P be a boundary pattern. Let

$$\mathcal{F} = (F_1, \dots, F_k)$$

be connected oriented surfaces properly embedded in M, in general position and transverse to P. Assume that the classes $[F_1], \dots, [F_k]$ are linearly independent in $H_2(M, \partial M; \mathbb{Q})$. Pairwise intersections are finite unions of arcs and circles. Triple intersections, if present, are included in the atlas; they do not split a component of a pairwise intersection.

Use the pattern complexity from [10, Section 3], measured against the fixed original pattern P:

$$c_P(F_i) = -4\chi(F_i) + |\partial F_i \cap P|, \qquad \iota(\mathcal{F}) = \sum_{1 \le i < j \le k} \#\pi_0(F_i \cap F_j). \tag{9}$$

Here $\#\pi_0$ counts geometric connected components, with no algebraic cancellation. The pattern in (9) is not the changing pattern of an intermediate cut.

Lemma 7.2 (Negative complexity and a nonzero relative class)**.** *Let F be a connected oriented surface properly embedded in M with $[F] \neq 0$ in $H_2(M, \partial M; \mathbb{Q})$. If $c_P(F) < 0$, then F is a violating disk for (M, P).*

Proof. A connected orientable surface of negative pattern complexity is either a sphere or a disk meeting P at most three times. An embedded sphere bounds a ball by irreducibility, so its relative homology class is zero. This contradicts the nonzero-class assumption. If a disk is not violating, the definition supplies a disk in ∂M with the same boundary and with the permitted empty, arc, or tripod intersection with P. Together the two disks bound a ball, after the usual small boundary perturbation, by irreducibility. The original disk is boundary-parallel and represents zero in relative homology, again contradicting the hypothesis. Thus the disk is violating. □

Accordingly, suppose throughout a nonterminal stage that every $c_P(F_i)$ is nonnegative. Define

$$Q(\mathcal{F}) = \sum_{i=1}^{k} c_P(F_i) + \iota(\mathcal{F}). \tag{10}$$

This is a nonnegative integer. The intersection term is essential: a loop can be essential in a cut-open face while bounding a disk in its uncut ambient surface. Such a loop leads to a disk exchange, rather than to a reduction in the surface's Euler characteristic.

Theorem 7.3 (Ordinary compression repair in an ambient block)**.** *Assume the preceding hypotheses and an authenticated atlas for $\mathcal{F}$. Suppose a disk D in one of its cut pieces satisfies the disk hypotheses of Lemma 7.1, and that its boundary α is essential in that cut-open face, which has label j. Then one can either obtain a violating disk for (M, P) or replace F_j by a connected oriented proper surface G, retaining all the other ambient surfaces, such that their k relative classes remain independent and*

$$Q(F_1, \dots, G, \dots, F_k) < Q(\mathcal{F}).$$

More precisely:

1. *If α is essential in the ambient surface F_j, then G can be chosen among the components of the compression of F_j along D, and in the nonterminal case*

$$c_P(G) \le c_P(F_j) - 4, \qquad \iota(\mathcal{F}_{\mathrm{new}}) \le \iota(\mathcal{F}).$$

2. *If α bounds a disk in the ambient F_j, then G can be obtained by a disk exchange. It has the same relative class, Euler characteristic, and boundary as F_j, while*

$$\iota(\mathcal{F}_{\mathrm{new}}) < \iota(\mathcal{F}).$$

Proof. By Lemma 7.1, regluing gives an embedded disk in M meeting F_j exactly in its boundary and avoiding every F_i with $i \neq j$. We continue to denote this image by D. A sufficiently small surgery neighborhood of D is disjoint from the other surfaces.

First suppose that α is essential in F_j. Let F_j' be the oriented surface obtained by compression. The surgery trace gives

$$[F_j'] = [F_j] \quad \text{in } H_2(M, \partial M; \mathbb{Q}). \tag{11}$$

Write V for the span of the other $k-1$ classes. At least one connected component G of F_j' has $[G] \notin V$, since otherwise their sum (11) would belong to V. Choosing such a component preserves rank k. In particular $[G] \neq 0$. If $c_P(G) < 0$, Lemma 7.2, with the newly independent family, supplies the asserted violating disk.

For the remaining case we prove the quantitative drop, including the effect of a possibly discarded component. Compression increases total Euler characteristic by two and leaves the boundary and its intersections with P unchanged. Consequently

$$c_P(F_j') = c_P(F_j) - 8. \tag{12}$$

There are at most two components. No component is a sphere. Indeed, a separating compression would produce a sphere only if α already bounded a disk in F_j, contrary to this case. A nonseparating compression produces a sphere only when F_j is a closed torus. Then (11) and irreducibility would force $[F_j] = 0$, again contrary to independence.

Every connected sphere-free orientable surface has Euler characteristic at most one, so a discarded component, if present, has pattern complexity at least -4. Subtracting that complexity from (12) yields $c_P(G) \le c_P(F_j) - 4$. All intersections with the other surfaces are disjoint from the surgery neighborhood. Each lies entirely in a component of the compressed surface, because it misses the compressed circle. Selecting G therefore deletes entire pairwise intersection arcs or circles and introduces none. This proves the first case.

Now suppose that α bounds a disk $E \subset F_j$. It must contain a pairwise intersection circle: if its interior missed every other surface, it would lift to a disk in the same cut-open face,

contradicting the essentiality hypothesis there. Since α is disjoint from the other surfaces and E lies in the interior of F_j, every pairwise intersection component meeting E is a whole circle contained in its interior.

The embedded sphere $E \cup D$ bounds a ball B by irreducibility. The connected surface $F_j \setminus \operatorname{int} E$ lies outside this ball. If it lay inside, then in the closed-surface case the whole F_j could be pushed into B, forcing its relative class to be zero. In the case with boundary, its boundary on ∂M rules out that possibility, since B is bounded by a sphere in the interior of M. Thus the usual disk exchange across B replaces E by a small parallel copy of D while retaining an embedded oriented proper surface. The isotopy may pass through other members of the family; their mutual disjointness during the isotopy is not a hypothesis.

The replacement disk is disjoint from all the other surfaces. It therefore deletes every intersection circle in E and creates no new intersection. At least one circle is deleted. The surface's relative class, Euler characteristic, and boundary are unchanged. This proves the second case, and hence the strict decrease of (10) in either nonterminal case. □

Corollary 7.4 (Additive bound for the specified repairs)**.** *Starting from one family satisfying the theorem, any sequence of its specified nonterminal repairs has length at most*

$$\left\lfloor \frac{C_0}{4} \right\rfloor + I_0, \qquad C_0 = \sum_i c_P(F_i^{\text{initial}}), \qquad I_0 = \iota(\mathcal{F}_{\text{initial}}).$$

This is at most $Q(\mathcal{F}_{\text{initial}})$. If the last repair instead terminates by producing a violating disk, the total number of repairs is at most $\lfloor C_0/4 \rfloor + I_0 + 1$.

Proof. The k classes remain independent. At each nonterminal stage their pattern complexities are nonnegative. Each nonterminal compression along a curve essential in its ambient surface reduces their sum by at least four; disk exchanges preserve that sum. There can therefore be at most $\lfloor C_0/4 \rfloor$ such compressions. Each disk exchange decreases the intersection count by at least one, and a compression never increases that count. There can consequently be at most I_0 disk exchanges. Adding these distinct operation counts proves the claimed bound. A terminal surgery is allowed to leave the nonnegative-complexity regime, so it is counted separately. □

Corollary 7.5 (Retaining individually nonzero labels)**.** *The geometric repair and additive bound remain valid if mutual independence is replaced by the weaker assumption*

$$[F_i] \neq 0 \quad \textit{for every } i.$$

In this version every retained label has a connected representative with nonzero relative class. No mutual independence conclusion is asserted. The number of labels is unrestricted by the dimension of the homology group. As before, the initial stage is nonterminal only if all its pattern complexities are nonnegative.

Proof. For an essential compression, the oriented sum of the classes of its components is the nonzero class $[F_j]$. Hence at least one component G has nonzero class. Choose that component. The exclusion of sphere components in the proof of Theorem 7.3 uses only $[F_j] \neq 0$ and irreducibility. The resulting estimate $c_P(G) \leq c_P(F_j) - 4$ is therefore unchanged. If $c_P(G) < 0$, Lemma 7.2 still produces a violating disk. Otherwise the same intersection argument proves a strict decrease of Q, and all labels still have nonzero classes.

In the disk-exchange case, the only homological input was that F_j cannot be pushed into a ball. Its nonzero class supplies exactly that input. The exchange preserves its individual class and strictly decreases the intersection term. The separate compression and disk-exchange counts in Corollary 7.4 are therefore unchanged: at most $\lfloor C_0/4 \rfloor + I_0$ nonterminal repairs and at most one additional terminal repair. □

Keeping every component of F'_j as one labeled oriented surface would preserve the individual class $[F_j]$. The connected-component selection used in the theorem preserves independence and the number of labels, but need not preserve that individual class. This distinction matters if a later construction requires one fixed cohomology basis rather than merely an independent family.

7.3 What the additive estimate does not establish

Ambient retention still requires updated cut data. The unchanged objects are the other surfaces embedded in M. Their pieces in the successive cuts, the induced boundary patterns, and their source certificates can change. Discarding one compressed component can glue together several old pieces of a later surface. The theorem therefore permits retaining ambient surface choices; it does not justify reusing old normal coordinates, handle structures, source digests, or suffix certificates without verification.

The disks have restricted boundary behavior. The hypotheses cover an ordinary compression whose boundary stays in one surface face and misses every seam. They do not cover general pattern-compressions involving arcs on the original boundary or several labeled faces. Such an extension must account for the changed pattern intersections and for the reconnection of seam traces. Nor does the theorem find all the disks it permits.

No simultaneous normalization bound is supplied. The repaired surfaces need not be normal in the previous handle structure. Normalizing one of them can change its intersections with the others. To use Corollary 7.4 inside a normal-surface algorithm, one would need to control that change or charge it to another bounded quantity. Likewise, irreducibility proves that the disk-exchange ball exists; this proof supplies no bound for a compressed description of the resulting isotopy.

The potential is geometric, not automatically polynomial in the input bit length. For example, take the two vertical annuli in $T^2 \times I$ above curves of slopes $(1, 0)$ and $(1, N)$ in minimal position. Their pattern complexities for $P = \varnothing$ are zero, but their pairwise intersection has N arcs. A slope or twist description can store N in binary. Thus even an additive bound in Q can be exponential in a compressed description length. The example illustrates the size of the bound; it does not assert that these annuli actually require N repairs.

Homological independence has limited force. The preserved assertions are embeddedness, orientation, properness, connectedness of each selected representative, and independence in the fixed ambient relative homology group. The theorem asserts neither incompressibility nor pattern-incompressibility. It also does not prove nonseparation of every later cut-open surface after several previous cuts, or the q-complexity descent required by the hierarchy algorithm.

For the independent version there is also a strict rank restriction: $k \leq \dim H_2(M, \partial M; \mathbb{Q}) = b_1(M)$ by Poincaré–Lefschetz duality. For a knot exterior, abelianizing a Wirtinger presentation identifies all arc generators with one meridian generator, so $b_1(M) = 1$. The independent version consequently cannot retain a whole multisurface hierarchy in the original knot exterior. Its multiple-label use requires an appropriate later ambient block of larger first Betti number and a suitable atlas there. Corollary 7.5 avoids this rank restriction by retaining only individual nonzero classes, and so permits several ambient labels even in a knot exterior. Neither version proves that the resulting pieces form a valid sequential hierarchy. In particular, the corollary does not replace the missing later-cut nonseparation argument.

The resulting research problem is concrete. Construct ambient atlases whose geometric intersection complexity is controlled, authenticate their seams without expanding binary multiplicities, and retain that control through simultaneous normalization and pattern-compression. The local cutting and bundle results, including the incoming relative-handle work [4], address different parts of this construction. The additive repair theorem does not close those remaining obligations or imply a global quasi-polynomial recognition bound.

8 A proposed research program

The next work should target proof obligations exposed by the construction, not merely increase the number of heuristic moves attempted. We propose the following program, arranged from the nearest implementable extension to the missing global complexity theorem. Each question includes a concrete outcome by which progress can be judged.

8.1 Immediate geometric and implementation questions

1. Can a canonical raw obstruction guide geometric normalization? The residual stopping set identifies precisely which original and saturated rows cannot disappear in the fixed model. For the eight retained native positive cuts with raw obstructions, locate a source-authenticated move that changes one of these rows' actual cocore intersections. Candidates include boundary bigon removal, a different meridian disc system, and a small certified change of the core handle structure. The outcome should be a before/after geometric certificate and a strictly smaller stated measure, rather than a freely reduced relator that happens to contain a singleton. This is a finite, already supplied test set on which a successful next theorem would have visible consequences.

2. How can geometric unit intersections be discovered efficiently? An algebraic exponent of one is insufficient. One needs an embedded attaching curve, an actual meridian disc, and a controlled isotopy or disc slide proving one geometric intersection. Lackenby's curve algorithms compute geometric intersection and isotopy information in polynomial time in surface triangulation size and logarithmic curve weights [11]. The research task is to connect their exact input and output structures to the source boundary charts here. A separate minimum-intersection calculation for each pair of curves would not by itself certify one simultaneous complete meridian system.

3. Which three-handle cores admit compressed cancellation? Seven of the ten retained three-handle cases are actual unions of handlebodies according to the explicit oracle. Develop a source verifier for a safe two/three-handle cancellation or a controlled retriangulation, including how every bundle port is transported. Start with these seven cases and retain the three negative controls. A sphere boundary or an Euler calculation must not authorize deletion of a closed summand. An eventual theorem should bound the number of new source records, not only the change in Euler characteristic.

4. Can private cocores be addressed without being enumerated? The present theorem supplies their count and initial incidence type, but not a name for every cocore in a genus-2^B bundle. A useful extension would provide a parameterized family of cocores with a bounded number of chart types, a binary address within each family, and independently checkable intersection queries. Its output contract should allow finite queries rather than require a literal list of all handles. The initial target can be patterns using only polynomially many source ports; the general case must allow the possibility of larger representation costs.

5. What boundary-pattern information survives the compression? An unmarked genus histogram omits cyclic order, endpoints, coorientation, and the placement of graph vertices. Design a richer certificate that retains this data through the collar construction, connector contractions, and geometric cancellations. Matching the genus and a matrix of unsigned intersections is not an adequate test. A strong validation target is an independent reconstruction of the induced pattern on small explicit cut boundaries, followed by exact agreement of essential-pattern tests.

6. Can the finite compiler be made linear in its explicit input? The new peeling pass is linear in incidence work, but the delivered relative component assembly and independent forest reconstruction are deliberately straightforward polynomial routines. Index each row, port, and base once by its component, and replace repeated scans by one grouped pass. Preserve the independent checker architecture rather than importing the producer's union–find routine. Measure the complete verified task: the current common-success timings show that optimizing a small finite subroutine may have little effect when orbit discovery dominates.

8.2 Compressed geometric transitions

7. Can the incoming cut grammars compile actual hierarchy moves? The incoming tile and first-return results already evaluate important compressed local queries[5, 6, 7]. The next theorem should start with authenticated source charts and end with the exact charts of the modified manifold, including the new boundary pattern. Each grammar edge needs a geometric carrier; each gluing needs orientation and endpoint checks. A polynomial evaluation bound for an already supplied grammar is only one part of this transition theorem.

8. When does a sequential hierarchy admit an ambient block atlas? Theorem 7.3 assumes an atlas; it does not construct one from arbitrary later surfaces. Characterize the obstruction to matching opposite boundary traces when those surfaces are reglued into an earlier manifold. An algorithm should either produce a properly embedded ambient family with verified local links, or exhibit the unmatched seam or self-intersection that prevents such a family. This makes the extra hypothesis testable rather than treating it as an informal picture.

9. Can many seam exchanges be batched with a real decrease bound? The sharpened repair estimate is $\lfloor C_0/4 \rfloor + I_0$, and I_0 can be exponentially large in a binary slope description. Seek a theorem that removes a whole certified parallel family of intersection circles in one compressed operation, with an amortized bound on the number of family changes. Merely storing I_0 in binary or taking its logarithm does not change the number of unit repairs. The output should include the new surface carriers and a proof that no remaining seam is reconnected incorrectly.

10. Can normalization respect the block potential? Ordinary repair leaves standard embedded surfaces that need not be normal in the current handle structure. Normalizing one while ignoring the others can increase intersection complexity. Prove either simultaneous normalization with controlled $\sum c_P + \iota$, or a second bounded quantity paying for every increase. The individually nonzero variant of the repair theorem allows many labels even in a knot exterior, but does not guarantee that every later cut-open surface is nonseparating. That sequential admissibility condition must be checked separately.

11. Can the component-selection step be performed from orbit data? The repair proof selects a compression component with nonzero relative class, or one outside the span of

the other classes. To implement that selection without expanding sheets, enrich the component oracle with authenticated relative homology weights and orientation data. Verify that the weights really add under regluing and that the selected component has the claimed boundary carrier. A vector of abstract homology labels without such carriers would not establish the embedded atlas hypotheses.

8.3 The genuinely global questions

12. Can useful surfaces be found within the intended candidate budget? A short certificate and a fast cut consumer do not bound discovery. Establish a coverage theorem for a specific family of normal or compressed surfaces: every eligible hierarchy state must contain a useful candidate within a stated, efficiently searchable family. The theorem must control both the number of alternatives and their encoded sizes. This is the place to connect the earlier sparse-support and tile kernels to the selection problem, rather than treating those kernels as an enumeration bound.

13. Is there a logarithmic-depth structural decomposition? To obtain the February exponent by Theorem 4.2, one sufficient target is $L = O(\log n)$ with polynomial candidate and state bounds. The March exponent permits $L = O((\log n)^2)$ under the same hypotheses. Seek a structural operation that decreases a suitable charged complexity by a fixed proportion, or an amortized substitute. A proof that a positive integer decreases by one gives termination and a linear numerical bound, which is not a logarithmic-depth theorem.

14. Can backward corrections be charged without resetting a full suffix? The ambient-block theorem demonstrates one sufficient local condition for retaining other surface choices. The next global result would have to decompose every relevant correction into such blocks, control the number and cost of block changes, and cover pattern-compressions and vertical-boundary moves. The goal is an honest replacement for the reset factor $(g+1)^L$, with a potential maintained across all those operations. Counterexamples in which a later surface cannot be reglued should be collected alongside successful block repairs.

15. Can all encoded-size bounds be composed from the diagram? Maintain a ledger for triangulation records, pattern records, normal coordinate bits, private-family descriptions, and certificate trace size after every operation. Prove a bound in the original crossing count, including transitions that simplify topology but enlarge its encoding. Only after discovery, transport, and reset costs fit one such ledger can the local results legitimately enter a global quasi-polynomial theorem. The final test is a proof of the hypotheses of Theorem 4.2, or a different complete cost argument, rather than an extrapolation of the benchmark curves.

8.4 Proposed order of work

The nearest next contribution is geometric normalization of the eight retained raw-positive failures, paired with a source-preserving two/three-handle rule for the seven retained three-handle positives. These tasks have concrete source cases, exact oracles at small scale, and a clear success criterion. Boundary-pattern transport and private cocore addressing should proceed alongside them, because an unmarked terminal result alone cannot drive the intended hierarchy. Ambient block construction and batched repairs then address the reset bottleneck. Candidate coverage and global size/depth control remain the final structural obligations.

A Reproduction and integration package

The accompanying archive includes this PDF and all LaTeX sources, the new producer and verifier modules, tests, raw source cases and certificates, the unchanged pinned runtime required by the examples, baseline comparison code, experiment scripts, and an additive integration patch. No remote repository branch has been changed by this continuation.

A.1 Minimal commands

From the extracted package root, run the standard-library extension tests:

```
PYTHONPATH=code:runtime python3 -B -m unittest discover -s tests -v
```

This includes all 106 precheck on/off comparisons and source replays. The recorded run used Python 3.12.14. The optional explicit-cut audit additionally needs Regina, recorded as version 7.4:

```
PYTHONPATH=code:runtime python3 -B experiments/native_audit.py
```

It repeats the old/new comparison and literal-cut oracle, retaining the normal-disc expansion cap. To replay the saved certificates without rerunning discovery or the oracle, use:

```
PYTHONPATH=code:runtime python3 -B experiments/replay_saved.py
```

Run the two benchmark scripts sequentially, with other experiment and test processes stopped:

```
python3 -B experiments/benchmark_column_peeling.py
PYTHONPATH=code:runtime python3 -B experiments/benchmark_native.py
```

The comparison scripts emit raw repetition arrays and A/A controls. Their results are local measurements and should be regenerated on the integration machine rather than treated as machine-independent constants.

To regenerate the static figure and build the article, use:

```
python3 -B experiments/generate_figures.py
python3 -B build_article.py
```

Figure generation uses Matplotlib. The supplied PNG and SVG are already present, so rebuilding the PDF itself requires only a normal LaTeX installation with the packages named in the main source. The build script runs `pdflatex` three times and preserves its log for inspection.

A.2 Important files

Path	Purpose
`code/column_peeling.py` and `code/column_peeling_verify.py`	Sparse matrix producer and independent timestamp/stopping-set verifier
`code/relative_bundles.py` and `code/relative_bundles_verify.py`	Finite geometric compiler and independent reconstruction
`code/normal_cut_saturation.py` and `code/normal_cut_saturation_verify.py`	Complete source-bound wrapper and replay, including optional early core refusal

Path	Purpose
`data/native_audit.json`	All 106 baseline/new results and actual Regina comparisons; no omitted refusals
`data/core_precheck_audit.json`	Enabled/disabled comparisons, identical success proofs, and exact query savings
`data/native_benchmark.json`	Complete-call timing repetitions and A/A controls
`results/column_peeling_validation.json` and `results/column_peeling_benchmark.json`	Algebraic validation counts, exact work counters, and raw kernel timings
`data/certificates/`	Seven new successes, legacy and binary controls, a full raw obstruction, and an early core obstruction
`data/extension_tests.log` and `data/native_regression_pinned.log`	Final 37-method extension run and 1,663-test pinned baseline run
`integration/additive_extension.patch`	Adds namespaced modules to the pinned repository without changing its default consumers
`INTEGRATION.md` and `CLAIMS.md`	Proposed integration sequence, proof obligations, and boundaries of the claims

The `runtime/fastunknot/` files and `baseline/core_handlebody_7046006.py` are unchanged excerpts of the inspected source. Their original licensing accompanies the package. The main implementation can be reviewed separately from those reference files. The observer-family lemma and ambient-block repair theorem are mathematical contributions only; no code in the package claims to solve their missing geometric input or transport problems.

A.3 A safe integration sequence

Apply the additive patch to a checkout containing the expected source interfaces, then import the new modules through the `fastunknot` namespace. Verify saved source certificates and run the added focused tests before putting the operator behind an explicit experimental option. Preserve the old full-row consumer while comparing census agreement and full-call costs. A later change to the default recognition schedule should state which selected surfaces reach this operator and how its unmarked census participates in the existing topological proof.

The present contribution is complete as an opt-in research operator and reproducible article. Its interface intentionally reports inconclusive cases that remain outside its hypotheses or fail its fixed geometric criterion. Those retained cases, not a claim of general quasi-polynomial recognition, are the basis for the next research stage.

References

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